Sagnik Bhattacharjee

Email | Twitter | LinkedIn | GitHub

Experience

Research Intern April 2026 – Present
Language Technologies Research Centre, IIIT Hyderabad *Hybrid*

- Co-authoring a paper on size agnostic DPPNets (in preparation), leading the empirical side of reproducing the Size-Agnostic DPPNet and VAE based pipelines and running the ablation, parameter reduction, and pruning experiments.
- Reverse engineered Hextimate, the hidden latency cost model inside Qualcomm's stripped Hexagon NPU compiler, recovering its class structure, bandwidth formula, and per chip coefficients from the binary. Writeup
- Built a batched Householder QR kernel in Triton for NVIDIA B200 GPUs, delivering a 9.3× speedup over PyTorch QR (4.79 ms geomean) via SRAM resident panel factorization. GitHub | Tweet
- Studied how different vLLM inference optimizations (prefix caching, chunked prefill, AWQ, admission control) perform across four traffic shapes, finding that several improve throughput while degrading per-user latency under sustained load. Writeup
- Building LensRT, a computation graph analyzer for LiteRT/QAIRT that maps absorbed operators to NPU partitions and explains rejection causes, enabling systematic diagnosis of compiler placement decisions.

Software Engineer Intern April – May 2026
Stealth *Remote*

- Built the data capture stage of a teacher-student **distillation pipeline** for **analog circuit optimization**, where an LLM agent edits SPICE parameters, runs **ngspice**, and logs trajectories for fine-tuning a smaller optimizer.
- Reduced **OPA317** optimization from non-convergence in 20 iterations to convergence in 4 by adding a **rolling history window** and an **in-memory PDK retrieval tool** over **SkyWater SPICE** libraries.
- Added **counterfactual** and **best-of-N** trace capture, turning optimization trajectories into **contrastive training pairs** for process-supervised fine-tuning.
- Built a **cross-file static analyzer** that validates SPICE netlists, testbenches, and metadata before simulation.

Projects

Spheni | Vector Search Library | *C++, Python, OpenMP* | GitHub January 2026

- Architected a vector search library from scratch with **Python bindings (pybind11)**, implementing Flat, IVF-Flat, PQ-Flat, and IVF-PQ indexes with **k-means++ clustering** and **inverted list storage**, achieving **12.6× speedup at 97% Recall@10** on 50k SIFT vectors via Voronoi-partitioned candidate scans.
- Implemented **Product Quantization** with residual encoding and **ADC** via precomputed sub-space lookup tables, compressing a 1M × 768-dim Cohere dataset from **2929 MB to 36.6 MB (80×)** at **83.3% Recall@10-in-100**.
- Eliminated **thread contention** in batch query execution via **thread-local Top-K heaps with OpenMP dynamic scheduling**, scaling throughput from **2.4k to 5.7k QPS (2.37×)** and cutting p99 latency from **0.554 ms to 0.287 ms (1.93×)**.

Skills

Languages: C++, C, Python
Systems: Linux, OpenMP, Git, Bazel, pybind11
ML Systems: vLLM, LiteRT, QAIRT, ONNX
Tools: Intel VTune Profiler, perf, GDB, ngspice, OpenBLAS

Education

Indian Institute of Engineering Science and Technology, Shibpur Kolkata, India
BTech in Information Technology *Aug 2025 – July 2029*